23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : 14

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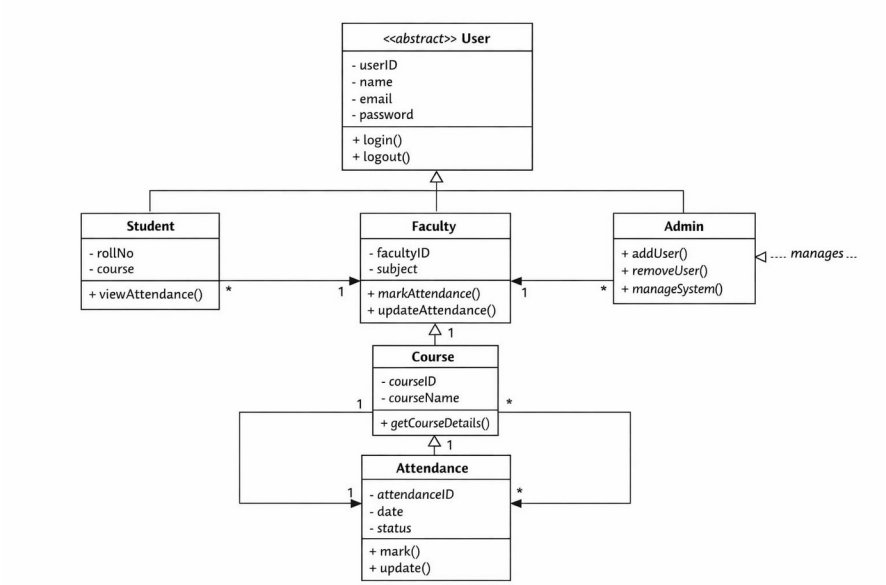
1. Problem Title : Attendance Management System

STRUCTURAL DESIGN

2. Class diagram



BEHAVIUORAL DESIGN

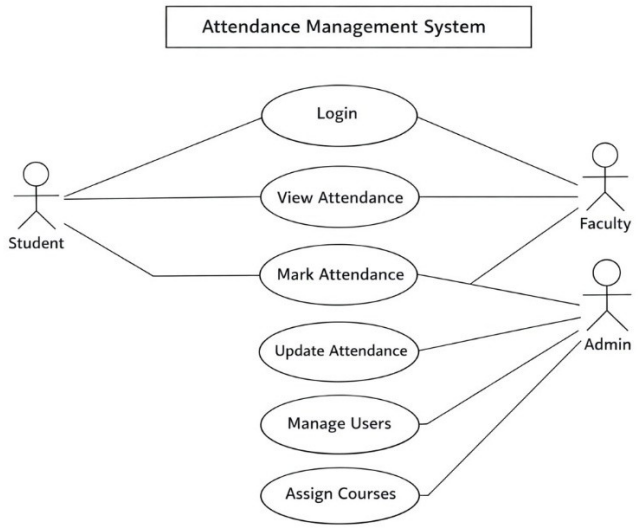
3. Use-case diagram
   a. List of actors
      1. Student

2. Faculty
3. Admin

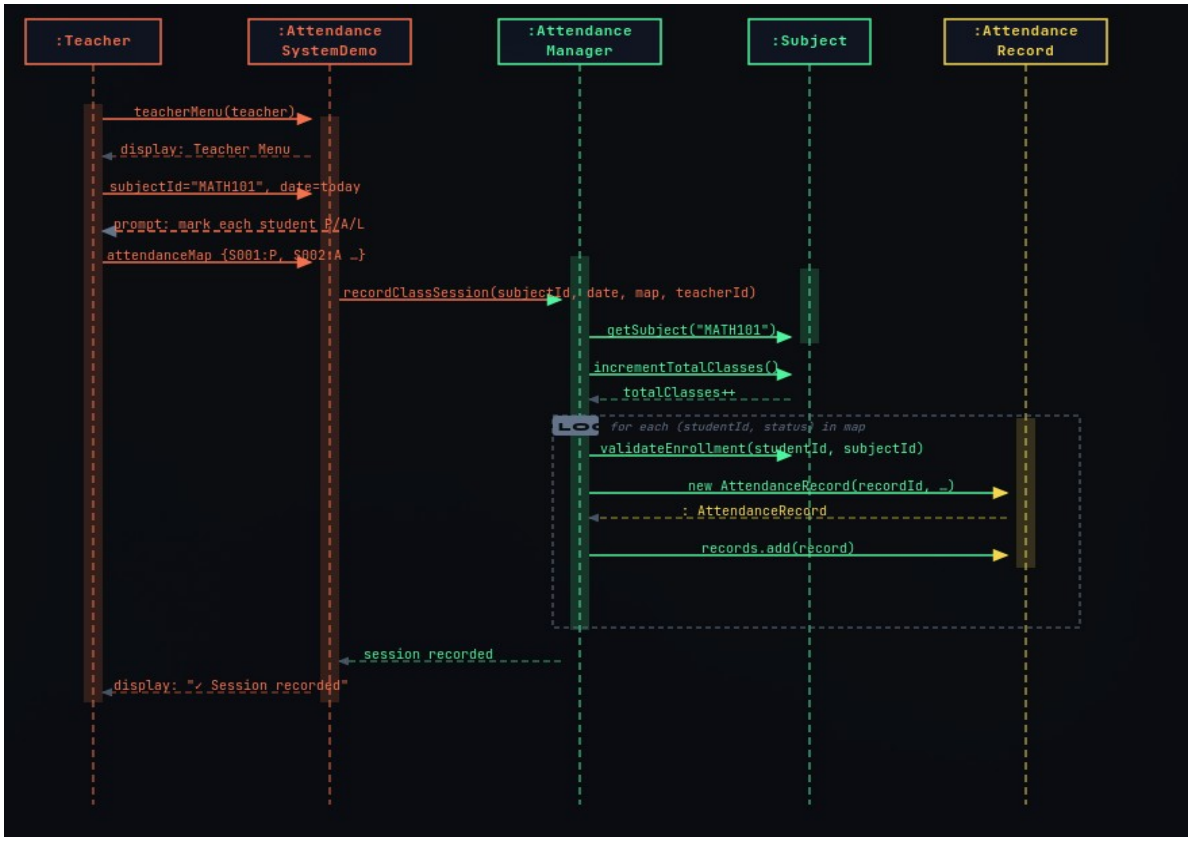
b. List of use-cases

1. Login :
   Users (both students and faculty) log into the system using credentials.
2. View Attendance :
   Students can check their attendance records.
3. Mark Attendance :
   Faculty marks attendance for students
4. Update Attendance :
   Faculty edits attendance if needed.
5. Manage Users :
   Admin adds/removes students and faculty.
6. Assign Courses :
   Admin assigns courses to faculty and students.

c. Use case diagram (use UML notations)

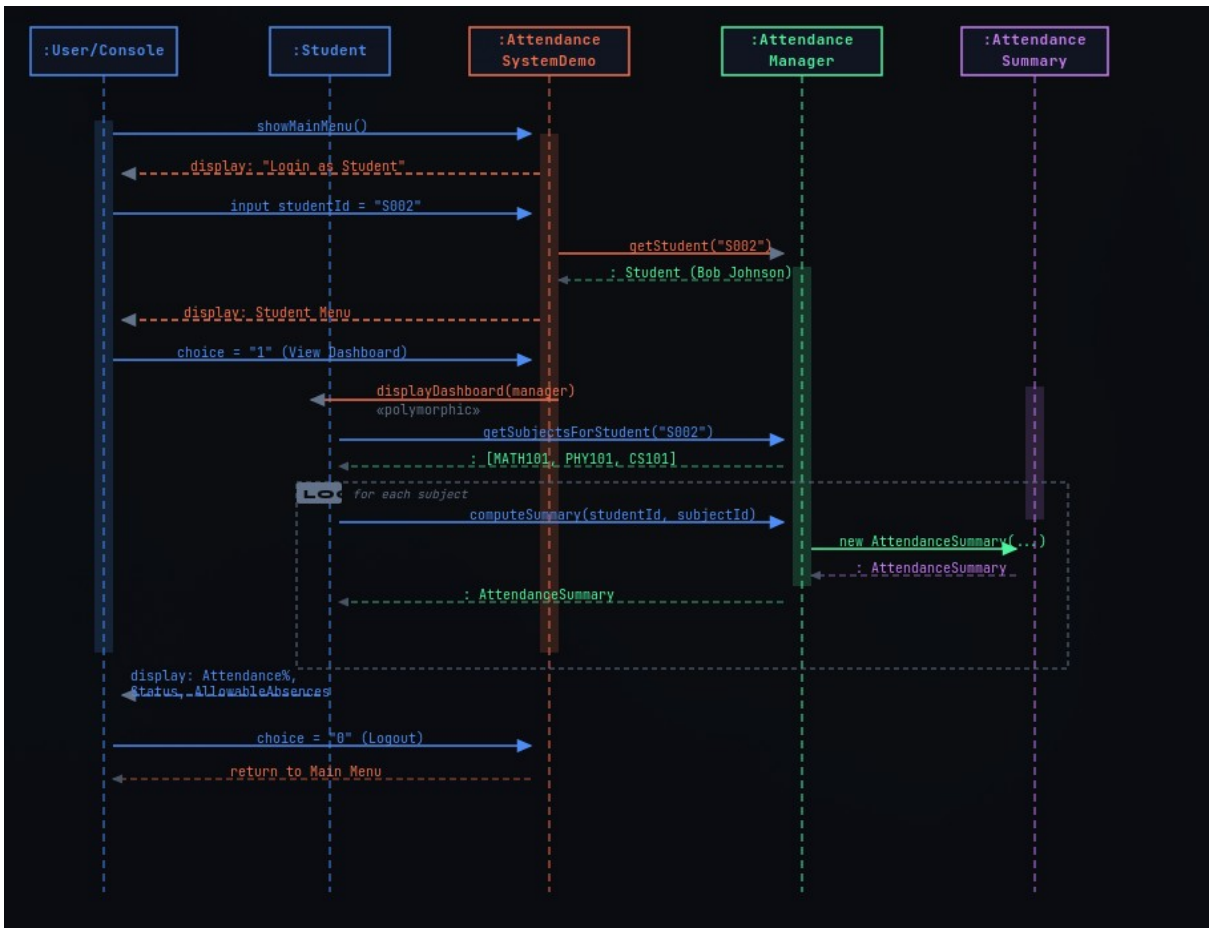


4. Sequence diagram (use UML notations)
   a. Teacher Records Attendance



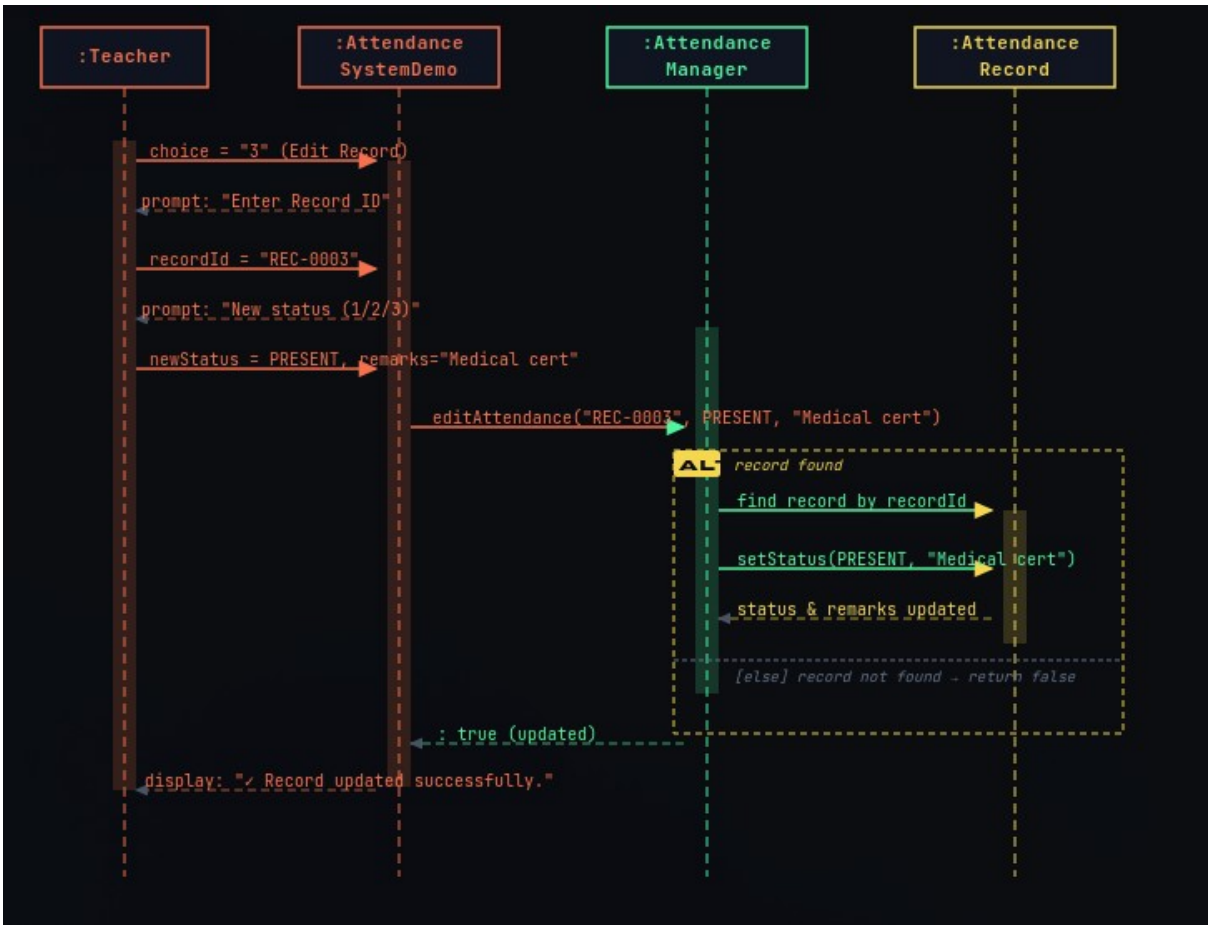
Scenario: The teacher selects a subject (e.g., MATH101) and inputs the attendance status for each enrolled student. The system validates the enrollment and creates new attendance records for the session.

b. Student Login & View Dashboard



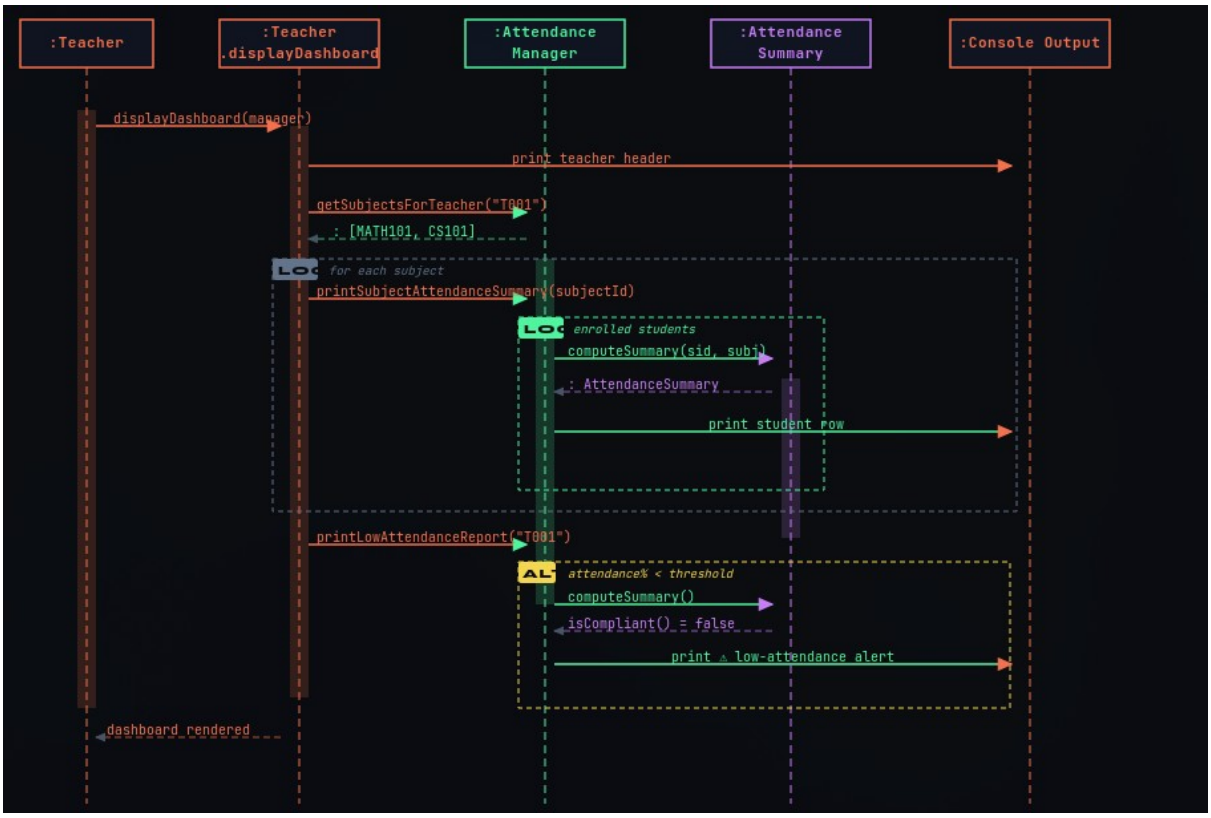
Scenario: A student logs into the console. The system fetches their enrolled subjects and computes an attendance summary for each, displaying their status and allowable absences.

c. Edit Attendance Record



Scenario: A teacher identifies an error in an existing attendance entry. They enter the specific Record ID and provide the updated status (e.g., changing from Absent to Present due to a medical certificate).

d. Teacher Dashboard & Alerts



Scenario: The teacher requests a dashboard view. The system iterates through the teacher's subjects, generates summaries for all enrolled students, and prints specific alerts for students whose attendance falls below the required threshold.